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$$KE = \frac{p^2}{2m}$$

Ans: **B**

2018 JC2 H2 Physics Prelims Exam Solution (**9749** Paper 1)

$$1.00 \times 10^6 \times 1.6 \times 10^{-19} = \frac{p^2}{2 \times 1.67 \times 10^{-27}}$$

$$p = 2.312 \times 10^{-20} \text{ kg m s}^{-1}$$

$$\Delta p = 0.01 \times 2.312 \times 10^{-20} = 2.312 \times 10^{-22}$$

$$\Delta p \Delta x = h$$

$$\Delta x = 2.87 \times 10^{-12} \text{ m}$$